Predicting League of Legends Match Outcomes from the First 15 Minutes

INTRODUCTION

For this year's project we decided to work with a League of Legends dataset. League of Legends is the most-played and most-watched esport in the world - a 5v5 team game in which two teams (Blue and Red) try to destroy each other's base, with professional matches lasting roughly 30 minutes. By the 15-minute mark, well before half of a typical game has elapsed, gold, experience and objective leads have already started to compound, and pro viewers often develop a strong feel for when a game is “already over”. Our project was to take real professional match data and find out how reliably this 15-minute snapshot forecasts the eventual winner, and whether machine learning can do better than the simple intuition the community already relies on.

THE DATA WE WORKED WITH

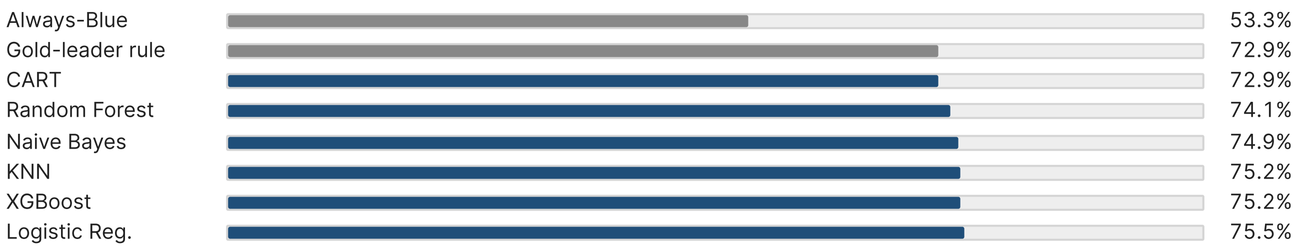
The main public source of professional match data is Oracle's Elixir, which collects every game played in every major regional league. We used the 2025 season export - 10,053 professional matches across 45 leagues (LPL, LCK, LEC, …). For each game it records detailed numbers for both teams and both map sides at the 10- and 15-minute marks: gold, experience, kills, farm and early objectives like first tower, first dragon and first blood. We reshaped the raw team rows into one row per match, removed mirrored columns and kept a focused set of 14 early-game features for the main models. We later tested 30 extra role-level features to check whether lane-specific information changed the result.

MODELS WE TRAINED

We trained six classifiers on the same 14-feature set (Logistic Regression, Random Forest, XGBoost, Naive Bayes, KNN, Decision Tree), and used three feature-selection methods (Forward Stepwise by AIC, LASSO, Elastic Net) to find which early-game variables still mattered after removing duplicated information.

WHAT WE FOUND

Machine learning helps, but only by a few percentage points. The naive rule “the team with more gold at minute 15 wins” already predicts 72.9% of matches on its own. Our best model (Logistic Regression) reaches 75.5%. A real improvement, but a modest one.



Test-set accuracy: baselines (grey) vs. trained models (blue).

We also tried pushing accuracy higher by adding per-role data: separate gold, experience and farm differentials for each lane (top, jungle, mid, bot, support) on top of the team totals. Bot lane was the strongest role-specific signal, but the improvement from adding all role features was small. In practice, the team-level totals already captured most of the useful information.

The feature-selection step was useful because it reduced the model without losing much accuracy. A compact seven-feature model kept almost the same predictive power as the full model: gold, experience and farm differences at 15 minutes, prior team win-rate difference, first tower, first herald and first dragon. Features such as first blood, plate difference and grub difference looked useful on their own, but did not add much once the main economic picture was already known.

Compared with published studies, our results are in the expected range. Earlier work on the same problem reports 76-78% accuracy at the 15-minute mark across model families and dataset types. Our 75.5% is close to those results, which suggests that the remaining errors come mostly from the unpredictability of the game after minute 15, not from a missing model.

CONCLUSION

Early-game numbers predict the winner fairly well, but not with certainty. LoL is a complex, fast-evolving team game in which many decisions after minute 15 still matter. Machine learning improves the gold-leader rule, but only slightly. Our best solution is not the largest model, but the compact seven-feature version: it is almost as accurate, easier to explain, and better suited for a Shiny tool where users need to understand why the prediction changes.